

In this Standard the comma has been adopted as the decimal sign

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NE 80-060

**ELECTRICAL BONDING - REPAIR OF COMPOSITE MATERIALS PARTS
ELECTRICALLY BONDED WITH CARBON FIBER FABRICS, EXPANDED
COPPER OR ALUMINUM FOIL AND BONDING STRIP**

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1. SCOPE

This Standard establishes the requirements and procedures for the repair of electrical bonding parts, such as carbon fiber fabric, carbon fiber bonding strip, expanded copper or aluminum foil and metallic bonding strip on composite material parts, in case of:.

- peeling off, fracture, displacement, delamination, damage or lack of expanded copper or aluminum foil or;
- peeling off, fracture, displacement, damage, lack or incorrect application of metallic bonding strip or carbon fiber bonding strip.

2. DEFINITIONS

2.1 CARBON FIBER FABRIC

Consists of a ply of carbon fiber fabric applied, among other applications, for electric bonding of composite parts. For carbon fiber reinforcement fabric and non crimped dry stacks, see MEP 15-048.

2.2 CARBON FIBER BONDING STRIP

Consists of a piece of carbon fiber fabric (2.1) applied for electrical bonding of composite parts.

2.3 EXPANDED COPPER OR ALUMINUM FOIL

Consists of an expanded foil to be applied on composite material laminates where electric conduction for electrical bonding is required. See MEP 02-009.

2.4 METALLIC BONDING STRIP

Consists of a piece of metallic sheet used to assure the adequate electric contact between components of the composite part and the expanded copper or aluminum foil. The metallic bonding strip is specified on MP 1.4.037 and can be made of:

- Phosphor bronze C51000 H02 (E7082204), according to ASTM B103;
- 2024 “0” aluminum clad alloy (E1010515), according to MEP 02-015 or Stainless Steel (CRES 300 Series), or
- Expanded copper foil (E3805391), Type II, Class 8, Grade 1, according to MEP 02-009.

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2.5 PROCESS QUALIFICATION

It is the evidence that this process is capable to turning parts to obtain the specified requirements and to guarantee that it is capable to reproduce the same results through the process stabilization. This can be reached controlling the key characteristics of the process.

3. APPLICABLE DOCUMENTS

Embraer Standards

MEP 02-009 - Lightning Wire Cloth and Expanded Foil.

MEP 02-015 - Aluminium Alloy Alclad 2024 , Fine Grained – Sheet.

MEP 06-025 - Nylon Film for Vacuum Bag.

MEP 09-063 - Structural Adhesive Film 121°C (250°F) and 177°C (350°F) Curing.

MEP 09-066 - Two Components Epoxy Paste and Liquid Adhesive for Bonding.

MEP 15-006 - Glass Fabrics Preimpregnated with Epoxy Resin 121°C (250°F)-Curing.

MEP 15-013 - Glassfiber Fabric Impregnated With Polytetrafluoroethylene (Teflon).

MEP 15-018 - Carbon Fabrics and Tape Preimpregnated with Resin Epoxy 121°C (250°F)-Curing.

MEP 15-048 - Carbon Fiber Reinforcement Fabric and Non Crimped Dry Stacks.

MP 1.4.037 - Sheet for Electrical Bonding of Composite Parts.

NE 20-038 - Repair in Composite Parts.

NE 40-006 - Aluminum and its Alloys - Chemical Conversion Coating.

PF 15-003 - Solid Laminate Curing to 121°C (250°F) - Manufacturing by Vacuum Process.

Foreign Standards

Federal Specification (GSA - USA)

O-E-760 - Ethyl Alcohol (Ethanol); Denatured Alcohol; Proprietary Solvents and Special Industrial Solvents.

AMS – Aerospace Material Specification- (SAE - USA)

SAE AMS-C-9084 - Cloth, Glass, Finished, for Resin Laminates.

ASTM – American Society for Testing and Material (USA)

B103 - Standard Specification for Phosphor Bronze Plate, Sheet, Strip, and Rolled Bar.

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4. MATERIALS AND EQUIPMENT

- 4.1 Milliohmmeter HP Model 4328A or equivalent.
- 4.2 Clean and dry white lint-free gloves.
- 4.3 Abrasive paper n°. 240 or finer.
- 4.4 Abrasive paper n°. 180 or finer.
- 4.5 Abrasive paper n°. 120 or finer.
- 4.6 Scotch Brite pad (type A)
- 4.7 Ethyl Alcohol, according to O-E-760, GR II, CLB (E 9120407)
- 4.8 Adhesive tape, glass fiber with PTFE (MEP15-013) (E 9122005).
- 4.9 Nylon Film for Vacuum Bag, according to MEP 06-025 (E2134437).
- 4.10 Expanded Copper Foil and Aluminum Wire Mesh Material, according to MEP 02-009 (E8056407).
- 4.11 EA9396 resin system, commercial (E6578982), MEP 09-066.
- 4.12 FEP Release Film or Demolding Film.
- 4.13 N-10 Cloth (E2139821).
- 4.14 Clamping plates.
- 4.15 Tongue depressor or squeegee.
- 4.16 Dry lint-free cloth.
- 4.17 Fiberglass Style 7781 (SAE AMS-C-9084).
- 4.18 Hot Bonder or equivalent.
- 4.19 Adhesive Film AF163, MEP 09-063.
- 4.20 Prepreg MEP 15-018 or MEP 15-006.
- 4.21 Bleeder (air weave's).
- 4.22 Electrical Bonding plate, MP 1.4.037.

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5. REQUIREMENTS

5.1 Repairs shall be made by qualified personnel having knowledge and experience in the repair of laminated parts.

5.2 Visual inspection shall be conducted to assure that the overall quality of the part is satisfactory, particularly in the area of repair to assure that the part is free from other damage.

5.3 All drawing requirements shall be met, such as weight limit, dimensional requirements and surface finish of the repaired area.

5.4 Parts that do not conform to drawing requirements after repair shall be resubmitted for repair or rejected.

6. PROCEDURES

This section describes repair of damage to areas of carbon fiber fabrics, expanded copper or aluminum foil using wet lay up (room temperature repair) and bonding strip repair using prepreg fabrics of high temperature. Damage to the carbon fiber fabric, copper or aluminum foil shall be designated as any defect sustained by a laminate in which the fabric/foil is exposed or broken (see Figure 1). The damages in the bonding strip are defined as disbonding and displacement of the plates.

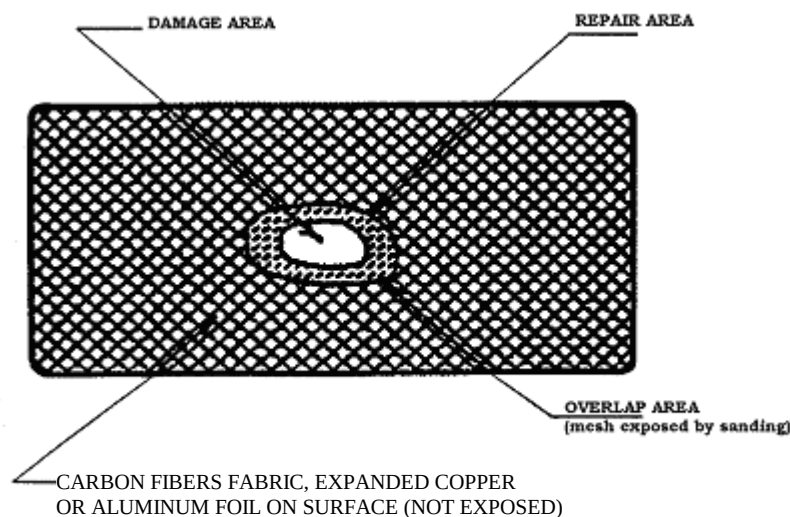


FIGURE 1: DAMAGE AREA

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6.1 EXPANDED COOPER OR ALUMINUM FOIL APPLICATION

Recommended milliohmmeter for resistance measurements is a HP Model 4328A or similar (4.1).

6.1.1 Processed parts shall be handled using clean white lint-free gloves (4.2).

6.1.2 Calculate the size of the repair area, that consists of the area of actual damage, (see Figure 2) and an overlapped area surrounding the damage area extending 25 mm (1 inch) out from the damage (see Figure 2).

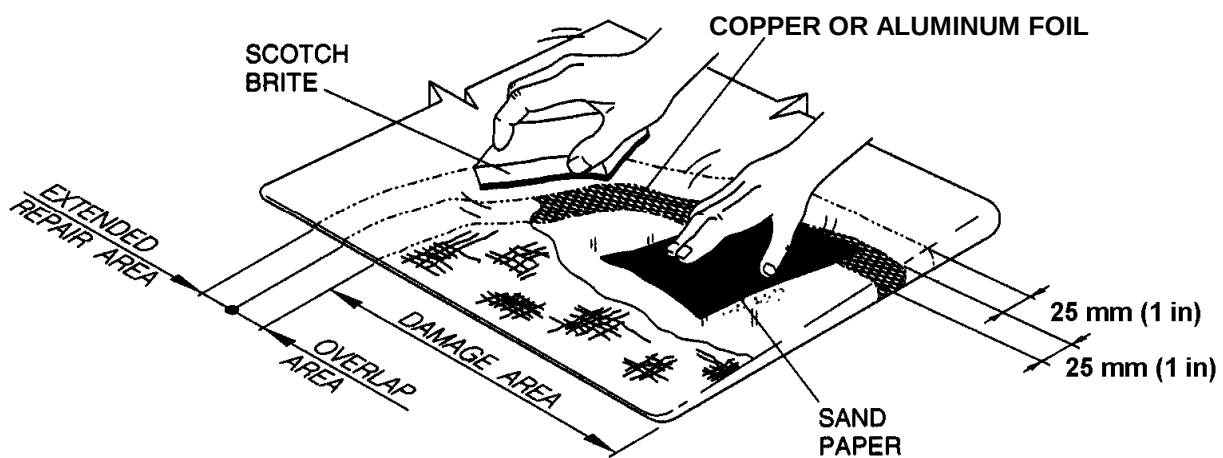


FIGURE 2: REPAIR AND OVERLAP AREAS

6.1.3 Using Number 180 or finer grit abrasive paper (4.4), lightly hand sand the 25 mm (1 inch) overlapped area enough to expose the expanded copper or aluminum foil (4.10 - see Figure 2). The fabric/foil is considered to be exposed when an electrical resistance measurement of less than 20 milliohms is obtained by lightly pressing the probe tips of the recommended milliohmmeter on the sanded overlap area. The probe tips shall be at least 25 mm (1 inch) apart. The reading shall be achieved in several places on the overlap area. The electrical measurements are for the purpose of assisting in the sanding process only.

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CAUTION

The expanded copper and aluminum foil are fine. Sanding down into the foil will cause damage to the fabric/foil and will result in measurements above 20 milliohms. Damage caused by oversanding is not visible to the eye and may occur without actually sanding through the foil. To prevent damage, use hand sanding or the number 180 or finer grit abrasive paper (4.4), on the rotary sander and make electrical resistance measurements frequently to evaluate the sanding progress. Stop sanding when the surface resistance is measuring less than 20 milliohms.

6.1.4 Using a fine “Scotch Brite” pad (Type A), lightly abrade the surface surrounding the repair area for an additional 25 mm (1 inch) in all possible directions. This is referred to as the extended repair area in Figure 2. The repair area, together with the extended repair area, makes up the entire repaired area. See Figure 3.

CAUTION

Do not expose the foil in the extended repair area.

6.1.5 Clean the repair area by saturating thoroughly with ethyl alcohol (4.7) using a clean, dry lint-free cloth (4.16). Wipe the surface with a second clean, dry, lint-free cloth prior to the evaporation of the solvent. Mask off around extended repair area with adhesive tape of glass fiber with PTFE (4.8). See Figure 3. Cover repair area with clean bagging film (4.9).

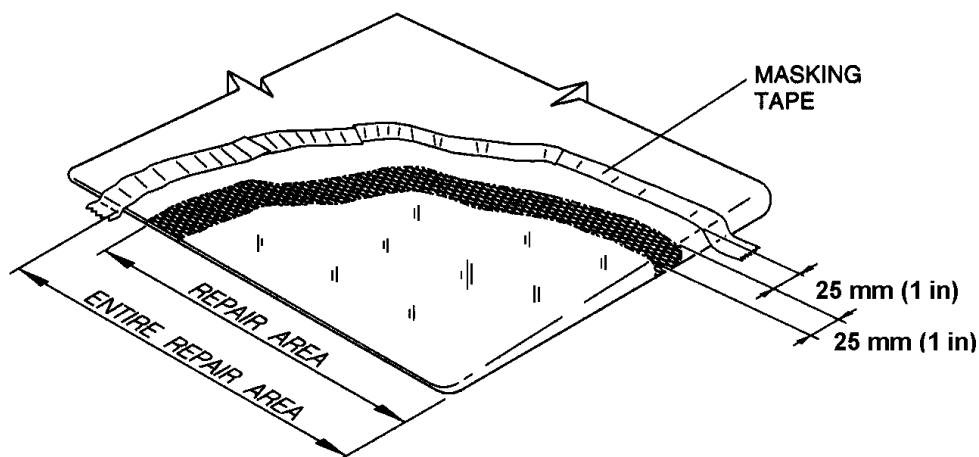


FIGURE 3: EXTENDED REPAIR AREA

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6.1.6 Select expanded copper or aluminum foil (4.10 - see Figure 2). Cut a piece of dry fabric/foil to size of entire repair area (considering an excedent area (25 mm – 1 inch, in all perimeter) for measurement purpose, see item 6.1.19) and solvent wipe with ethyl alcohol. To obtain a smoother surface in areas of complex curvature, the foil patch may consist of several overlapping pieces, provided that each piece makes contact with the overlap area.

6.1.7 Mix EA9396 resin system (Mix Ratio A/B = 100/30).

6.1.8 Remove bagging film from repair area and apply a thin coat of EA9396 to the entire repair area surface. See Figure 4.

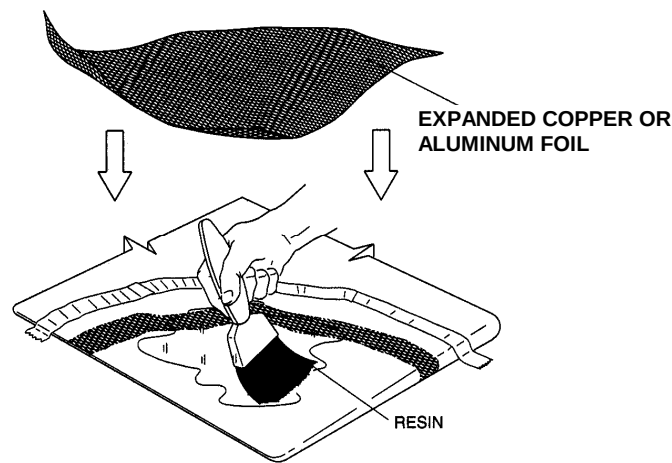


FIGURE 4: EXPANDED COPPER OR ALUMINUM FOIL

6.1.9 Place the repair foil over the repair area and apply a thin coat of EA9396 to the top surface of the foil.

6.1.10 Using a tongue depressor or squeegee, carefully remove excess resin from underneath the replacement expanded copper or aluminum foil.

6.1.11 Cut a piece of FEP (4.12) to the size of the repair area. See Figure 2. Place FEP on repair area.

6.1.12 Cut a piece of N-10 synthetic felt cloth (4.13) to extend past the extended repair area and place on repair.

6.1.13 Cover with nylon bagging film (4.9).

6.1.14 Apply pressure to the repair using a vacuum bag or clamping plates to keep the wet patch stationary.

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CAUTION

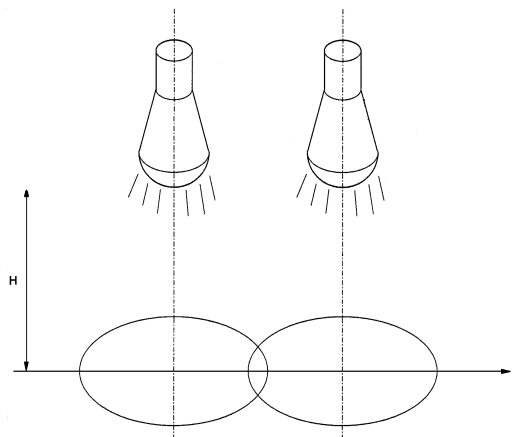
This is a critical step. If insufficient pressure is applied, the electrical measurements will be high. If mechanical pressure is applied with clamping plates, it should be evenly distributed and equivalent to the pressure typically applied by vacuum 68,9 kPa to 103,4 kPa (10 psi to 15 psi).

6.1.15 Cure cycle is $25^{\circ}\text{C} \pm 5,5^{\circ}\text{C}$ ($77^{\circ}\text{F} \pm 10^{\circ}\text{F}$) for 3 to 5 days or $65,5^{\circ}\text{C} \pm 5,5^{\circ}\text{C}$ ($150^{\circ}\text{F} \pm 10^{\circ}\text{F}$) for 60 minutes (minimum). For cure at $65,5^{\circ}\text{C} \pm 5,5^{\circ}\text{C}$ ($150^{\circ}\text{F} \pm 10^{\circ}\text{F}$), a thermal blanket, a curing oven, or heat lamps can be used.

NOTES

1) If heat lamps are used as a heat source to accelerate the cure, a variation of $3,9^{\circ}\text{C}$ (7°F) per minute is allowed, and the lamp height above the parts shall be according to Figure 5.

2) Curing time does not include the time required for the mold and part to heat up to the proper temperature. Curing time is the period after the part has reached that temperature.



TEMPERATURE °C (°F)	HEAT LAMP HEIGHT (INFRA-RED LAMP (250W) (1))	
	mm	(in)
57 (135)	381,0	(15,0)
60 (140)	345,4	(13,6)
66 (150)	309,9	(12,2)
71 (160)	284,5	(11,2)
77 (170)	266,7	(10,5)
82 (180)	248,9	(9,8)
94 (190)	233,4	(9,2)
93 (200)	223,5	(8,8)

For lamps different from 250W, to use a thermocouple in the part to monitor the temperature, redefining the distance " H ".

FIGURE 5: HEAT LAMP HEIGHT AND PART TEMPERATURE

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6.1.16 Remove the peel ply cloth from the repair area.

6.1.17 If necessary, lightly hand sand the entire repair area using Number 240 or finer grit abrasive paper (4.3) to remove any surface imperfections caused by resin squeeze out. Remove all dust before vacuuming part or wiping with clean, dry, lint-free cloth (4.16).

CAUTION

Do not expose the carbon expanded copper or aluminum foil when sanding.

6.1.18 Using the milliohm meter (4.1) specified in 6.1, electrical resistance measurements shall be taken between the expanded copper or aluminum foil that was used to repair the affected area and the adjacent embedded foil. Maximum resistance shall be 40 milliohms. This measurement shall be verified by Quality Assurance.

6.1.19 The adjacent foil may have to be exposed by sanding to make the measurements. Use the same sanding precautions as called out in 6.1.3. After the measurement is made, cut out the excedent area (see 6.1.6), and cover the exposed foil area with a light coat of EA9396 laminating resin. Mix and cure in accordance with items 6.1.7 and 6.1.15.

6.1.20 If the foil in the extended repair area is damaged, repeat the procedure, except that the repair area shall now include the damaged foil.

6.2 METALLIC OR CARBON FIBER BONDING STRIP APPLICATION

NOTES

1) Phosphor bronze or CRES strips shall be sanded (150 grit) and degreased prior to the bonding. For aluminum strips, the surface shall be treated by chemical conversion applying Alodine 1200 according to NE 40-006 Type I or III.

2) For the electrical bonding of composite parts, it is recommended the expanded copper foil strip (E3805391) application as bonding strip. Other materials can be applied after approval of engineering (EMI/Lightning and Materials Engineering)

3) In case of fixation of bonding strips in points where holes go through the bonding strips, it is allowed the strips displacement since 80% of the prescribed holes through the strips maintain entirely inside its contour and since the strip do not displace to the finished external border of the composite part.

4) For windows and access doors contours of parts manufactured with carbon fibers that apply more than 10 fasteners in the assembly of the window/access door, it is not necessary the use of metallic strips along the contour.

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6.2.1 WET LAY UP REPAIR

6.2.1.1 Inspect part to be reworked to ensure thickness of repair will not interfere with mating parts on next assembly.

6.2.1.2 Processed parts shall be handled using clean white lint-free gloves (4.2).

6.2.1.3 Calculate the size of the area to be repaired that is the damage area (see Figures 2 and 3) plus a overlap area around the damage area exceeding 25 mm (1 inch) beside the damage. See Figure 6.

6.2.1.4 Remove the strip and the lay-up on the plate using Number 120 or finer grit abrasive paper (4.5), lightly hand sand the area enough to expose the expanded copper or aluminum foil (see Figure 2). The expanded copper or aluminum foil is considered to be exposed when an electrical resistance measurement of less than 20 milliohms can be achieved by lightly pressing the probe tips of the recommended milliohmmeter on the sanded area. The reading shall be achieved in several places on the sanded overlap area. The electrical measurements are for the purpose of assisting in the sanding process only and Quality Assurance verification is not required at this point.

CAUTION

The expanded copper or aluminum foil is fine. Sanding down into the foil will cause damage to the expanded foils and will result in high resistance measurements. Damage caused by oversanding is not visible to the eye and may occur without actually sanding through the foil. To prevent damage, use hand sanding or the number 120 or finer grit abrasive paper (4.5), on the rotary sander and make electrical resistance measurements frequently to evaluate the sanding progress. Stop sanding when the surface resistance is measuring less than 20 milliohms.

6.2.1.5 Clean the repair area and the back of the bonding strip by saturating thoroughly with ethyl alcohol (4.7) using a clean, dry lint-free cloth (4.16). Wipe dry with a second clean, dry, lint-free cloth prior to the evaporation of the solvent. Mask off around repair area with adhesive tape of glass fiber with PTFE (4.8). Cover repair area with clean bagging film (4.9).

6.2.2 COLD REPAIR OF BONDING STRIP

6.2.2.1 Mix EA9396 resin system (Mix Ratio A/B = 100/30).

6.2.2.2 Remove bagging film from repair area and apply a thin coat of EA9396 to the entire repair area surface.

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6.2.2.3 Place the bonding strip over the repair area. Place one or two fiberglass ply style 7781 or carbon plain weave fabric with EA9396 (4.11), overlapping 13 mm (0,5 in) from the strip and apply a thin coat of EA9396 to the top surface of the strip. See Figure 6.

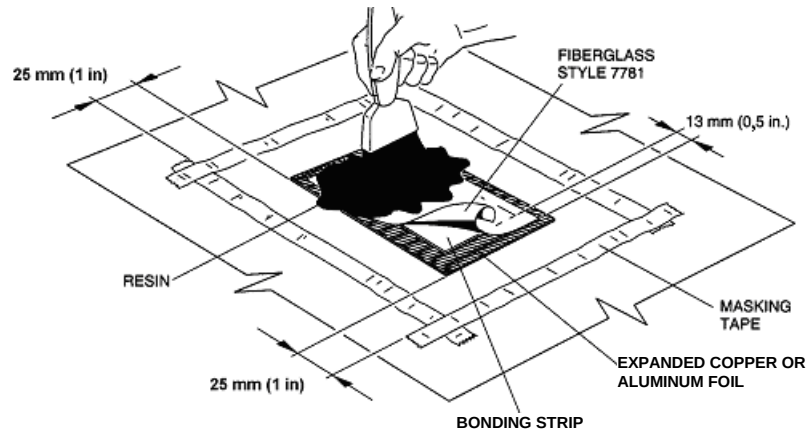


FIGURE 6: BONDING STRIP

6.2.2.4 Using a tongue depressor or squeegee, carefully remove excess resin from underneath the replacement bonding strip.

6.2.2.5 Calculate the size of the area to be repaired that is the damage area (see Figure 6) plus a overlap area around the damage area exceeding 25 mm (1 inch) beside the damage.

6.2.2.6 Cut a piece of FEP (4.12) to the size of the repair area. See Figure 7. Place FEP on repair area.

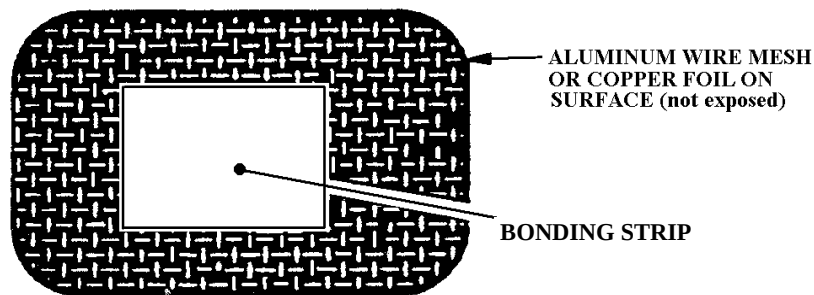


FIGURE 7: REPAIR AREA

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6.2.2.7 Cut a piece of N-10 synthetic felt cloth (4.13) to extend past the extended repair area and place on repair.

6.2.2.8 Cover with nylon bagging film (4.9).

6.2.2.9 Apply pressure to the repair using a vacuum bag or clamping plates to keep the wet patch stationary.

CAUTION

This is a critical step. If insufficient pressure is applied, the electrical measurements will be high. If mechanical pressure is applied with clamping plates, it shall be evenly distributed and equivalent to the pressure typically applied by vacuum 68,9 kPa to 103,4 kPa (10 psi to 15 psi).

6.2.2.10 Cure cycle is $25^{\circ}\text{C} \pm 5,5^{\circ}\text{C}$ ($77^{\circ}\text{F} \pm 10^{\circ}\text{F}$) for 3 to 5 days or $65,5^{\circ}\text{C} \pm 5,5^{\circ}\text{C}$ ($150^{\circ}\text{F} \pm 10^{\circ}\text{F}$) for 60 minutes (minimum). For cure at $65,5^{\circ}\text{C} \pm 5,5^{\circ}\text{C}$ ($150^{\circ}\text{F} \pm 10^{\circ}\text{F}$), a thermal blanket, a curing oven or heat lamps can be used.

NOTES

1) If heat lamps are used as a heat source to accelerate the curve, a variation of $3,9^{\circ}\text{C}$ (7°F) per minute is allowed and the lamp height above the parts shall be according to Figure 5.

2) Curing time does not include the time required for the mold and part to heat up to the proper temperature. Curing time is the period after the part has reached that temperature.

6.2.2.11 Remove the peel ply cloth from the repair area.

6.2.2.12 If necessary, lightly hand sand the entire repair area using Number 240 or finer grit abrasive paper (4.3) to remove any surface imperfections caused by resin squeeze out. Remove all dust before vacuuming part or wiping with clean, dry, lint-free cloth (4.16).

CAUTION

DO NOT EXPOSE THE EXPANDED COPPER OR ALUMINUM FOIL WHEN SANDING.

6.2.2.13 Using the milliohmmeter (4.1) specified in 6.1, electrical resistance measurements shall be taken between the replacement bonding strip and the adjacent embedded foil. Maximum resistance shall be 40 milliohms. This measurement shall be verified by Quality Assurance.

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6.2.2.14 The bonding strip may have to be exposed by sanding to make the measurements. Use the same sanding precautions as called out in 6.1.3. After the measurement is made cover the bonding strip and exposed area with a light coat of EA9396 laminating resin. Mix and cure in accordance with items 6.1.7 and 6.1.15.

6.2.2.15 If the expanded copper or aluminum foil in the extended repair area is damaged, repeat the procedure, except that the repair area shall now include the damaged copper or aluminum foil.

6.2.3 REPAIR USING HIGH CURE TEMPERATURE (120°C) MATERIALS

6.2.3.1 Prepare the prepreg (Table 1) and adhesive film (Table 1) and expanded copper or aluminum foil (4.10) in an enough quantity to perform the repair.

6.2.3.2 Remove the mold release film (non porous) from the repair area and apply an adhesive film (Table 1) to the repair area not covering a whole area, according indicated in the Figure 8.

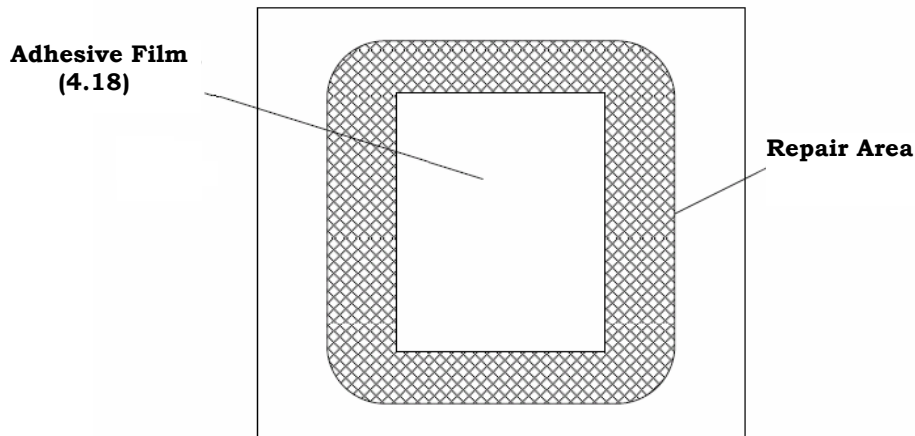


FIGURE 8: ADHESIVE FILM PLACED ON REPAIR AREA

6.2.3.3 For the bonding strip, use Phosphor Bronze 5% Sn or Stainless Steel (CRES 300 Series), as alternative. Aluminum shall be avoided due corrosion purposes, use it only if required. Direct contact of bronze or CRES bonding strip with aluminum structures are not allowed.

NOTES

1) Make sure that the bonding strip be correctly prepared to bonding (see Note of item 6.2).

2) If necessary, replace the bonding strip with new one. Make sure that the thickness of the bonding strip is the same as that of the original one.

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6.2.3.4 Place the bonding strip over the first adhesive film ply placed.

6.2.3.5 Place an expanded copper or aluminum foil (4.10) over the bonding strip overlapping about 12,7mm beyond of the whole contour of the first adhesive film ply (Table 1), assuring the direct contact among the inserted metallic foil and the existing metallic foil for, at least, 25,4 mm.

NOTE

If part of the existing metallic foil has been uncrated due to the damage, it shall be assured that the inserted metallic foil is in direct contact with the part of the existing metallic foil that has not been uncrated for, at least, 25,4 mm, according to Figure 9.

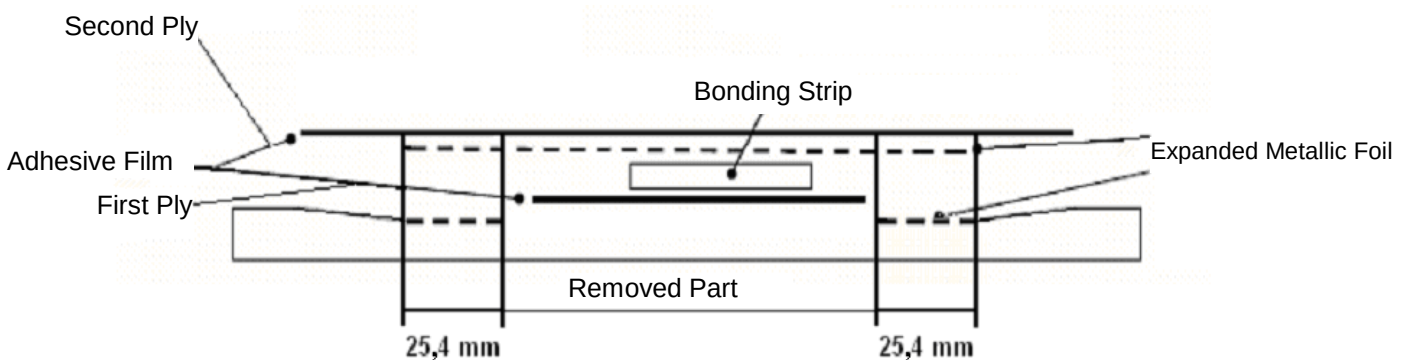


FIGURE 9: APPLICATION OF THE METALLIC FOIL

6.2.3.6 Place a second adhesive film ply (Table 1) over the bonding strip in the entire repair area. Refer to Figure 10.

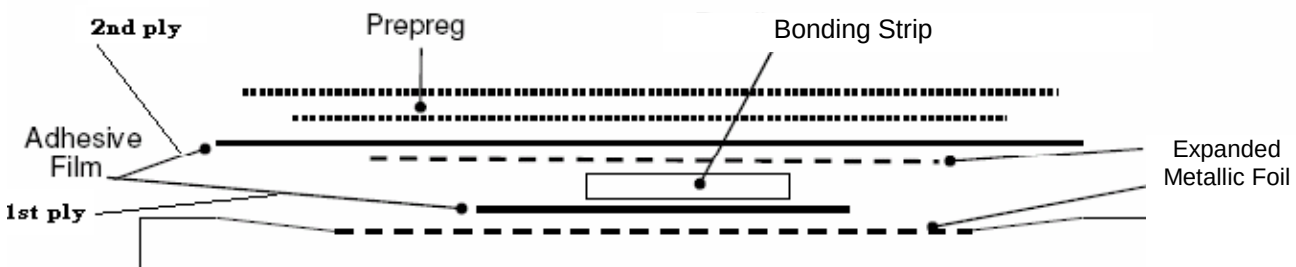


FIGURE 10: LAY-UP OF REPAIR

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6.2.3.7 Remove and recompose lay up on the bonding strip according to design using Table 1 materials.

TABLE 1: HOT REPAIR – USED MATERIALS

ORIGINAL MATERIAL	PN	REPAIR MATERIAL	ADESIVE FILM	REPAIR TEMPERATURE °C	PROCEDURE
MEP 15-006 (E1910496)	7781-38-F155-117 (L5765)	E1910496	MEP 09-063 (E2086261)	121	"HOT BONDER" / VACUUM / OR OVEN
MEP 15-006 (E1910507)	116-38-F155-116 (L5926)	E1910507	MEP 09-063 (E2086261)	121	"HOT BONDER" / VACUUM / OR OVEN
MEP 15-026 (E8353578)	HT93-7781 (B021033)	REPAIR NOT APPLICABLE TO THIS STANDARD OPEN NO CONFORMITY DOCUMENT			
MEP 15-018 (E1390026)	F3X-584-42-F155-113 (L6831)	E1390026	MEP 09-063 (E2086261)	121	"HOT BONDER" / VACUUM / OR OVEN
MEP 15-018 (E3157120)	L5821ED	E3157120	MEP 09-063 (E2086261)	121	"HOT BONDER" / VACUUM / OR OVEN
MEP 15-008 (E1910513)	7781-38-F161-222 (P1178)	E1910496	MEP 09-063 (E2086261)	121	"HOT BONDER" / VACUUM / OR OVEN
MEP 15-016 (E9125613)	7781-38-F120-219 (B021032)	E9125613	DO NOT USE	177	AUTOCLAVE / VÁCUUM / 100 psi
MEP 15-022 (E2070720)	F4G-584-F584 (P1140)	E1390026	MEP 09-063 (E2086261)	121	"HOT BONDER" / VACUUM / OR OVEN
MEP 15-022 (E2192954)	F4G-584-F584-RC34 (P1157)	E1390026	MEP 09-063 (E2086261)	121	"HOT BONDER" / VACUUM / OR ESTUFA

NOTES

1) If the repair consists of removing a bonding strip placed at the wrong place, after its removal, the lay-up shall be recomposed and the gap shall be filled with the materials according to Table 1.

2) In case composite repair material is part of the hot repair procedure adhesive film shall be cured using cure cycle applicable for that hot repair material. For procedures where only adhesive film specified by MEP 09-063 is applicable, the maximum time at the specified temperature shall be 90 minutes.

6.2.3.8 Cut a layer of bleeder so that the edges extend beyond the edge of the mold release film (porous).

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6.2.3.9 Cover the repair area with vacuum bag film.

6.2.3.10 Apply vacuum to the region.

NOTE

If the repair region allows it, you can use clamping plates to apply pressure. In this case, pressure shall be evenly distributed and equivalent to the pressure typically applied by vacuum. If insufficient pressure is applied, the electrical measurements will be high.

WARNING

USE HEAT CURING EQUIPMENT THAT IS ACCEPTABLE TO LOCAL FIRE PROTECTION AUTHORITIES. FAILURE TO COMPLY COULD CAUSE INJURY TO PERSONNEL.

6.2.3.11 Cure the repair at 120°C under vacuum pressure (using a hot bonder).

CAUTION

DO NOT EXPOSE THE PART TO TEMPERATURE OVER 120°C.

6.2.3.12 After the cure, remove the vacuum bag and lightly hand-sand the entire repair area to remove any surface imperfections caused by resin squeezing out. Use a number 240 or finer grit abrasive paper (4.3).

CAUTION

DO NOT SAND THE EXPANDED COPPER OR ALUMINUM FOIL.

6.2.3.13 Remove all dust from the sanded area by wiping the area dry with a clean, dry, lint-free wiper cloth.

6.2.3.14 Using the milliohm meter (4.1), electrical resistance measurements shall be taken between the replaced bonding strip and the adjacent embedded foil. Maximum resistance shall be 40 milliohms. This measurement shall be verified by Quality Assurance.

6.3 CARBON FIBER FABRIC APPLICATION

The repair of damage to areas of carbon fiber fabrics shall be performed according to NE20-038 (item Breaking of Ply and Delamination).

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7. REMARKS

7.1 This Standard shall be referenced on the material rejection report of composite materials parts, through the following documents:

- a) “CD - COMUNICADO DE DISCREPÂNCIAS”, for all programs “EMBRAER ;
- b) DR - DISCREPANCY RECORD, for programs S-92A and H92.

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